

Football Match Manager - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager project aims to create a system for managing football matches, with a focus on data modeling, API design, and security.

1.2 Maturity Assessment

The project is currently in the REFINEMENT phase, with a health index of 80.0 and a completeness score of 100.0. However, there are still some structural gaps that need to be addressed, including the addition of sections for data models and schemas, API contracts and flow, and security and identity.

1.3 Technical Stack

- Python 3.11
- FastAPI
- Relational database management system
- API gateway
- Identity and access management system
- Performance monitoring and optimization tools
- Security testing and vulnerability assessment tools

1.4 Architectural Constraints

- Relational data model
- API design with contracts and flow
- Security measures with identity management
- Performance goals with a target of 1000 req/s
- Constraints with a maximum response time of 200ms

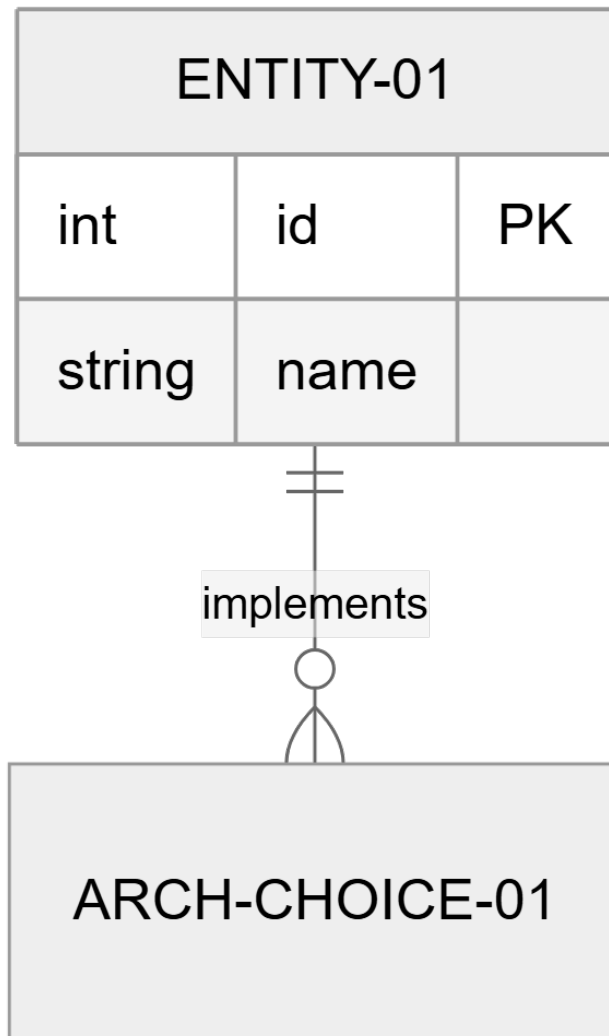
1.5 Critical Dependencies

- Relational database management system
- API gateway
- Identity and access management system
- Performance monitoring and optimization tools
- Security testing and vulnerability assessment tools

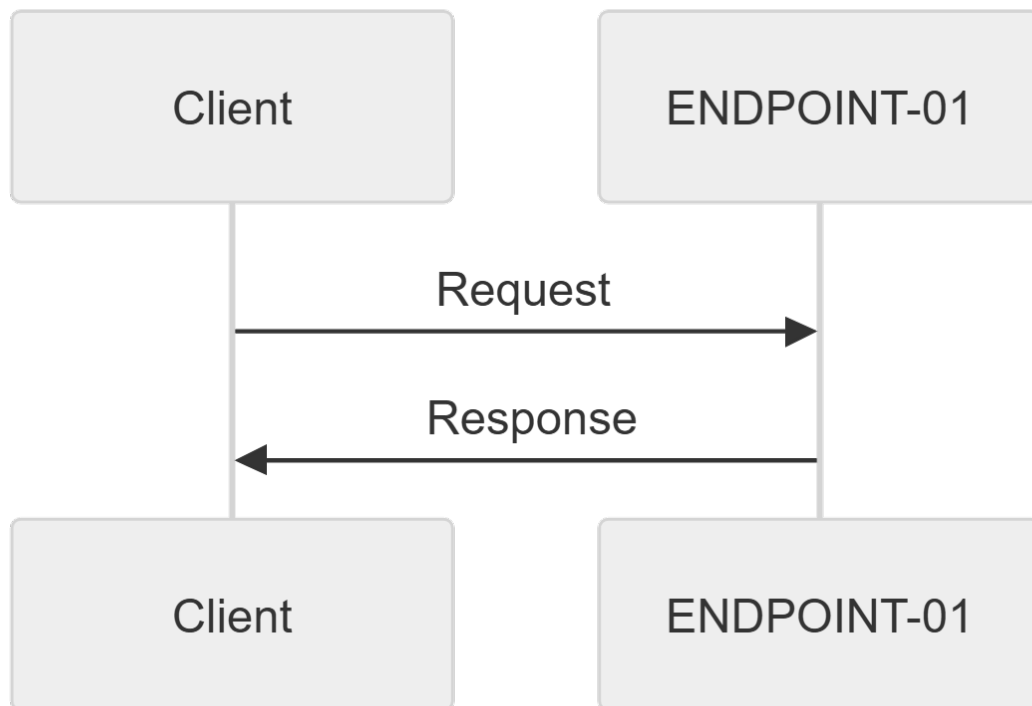
2. Architecture Workflows & Visual Diagrams

The system's architecture can be visualized through the following diagrams:

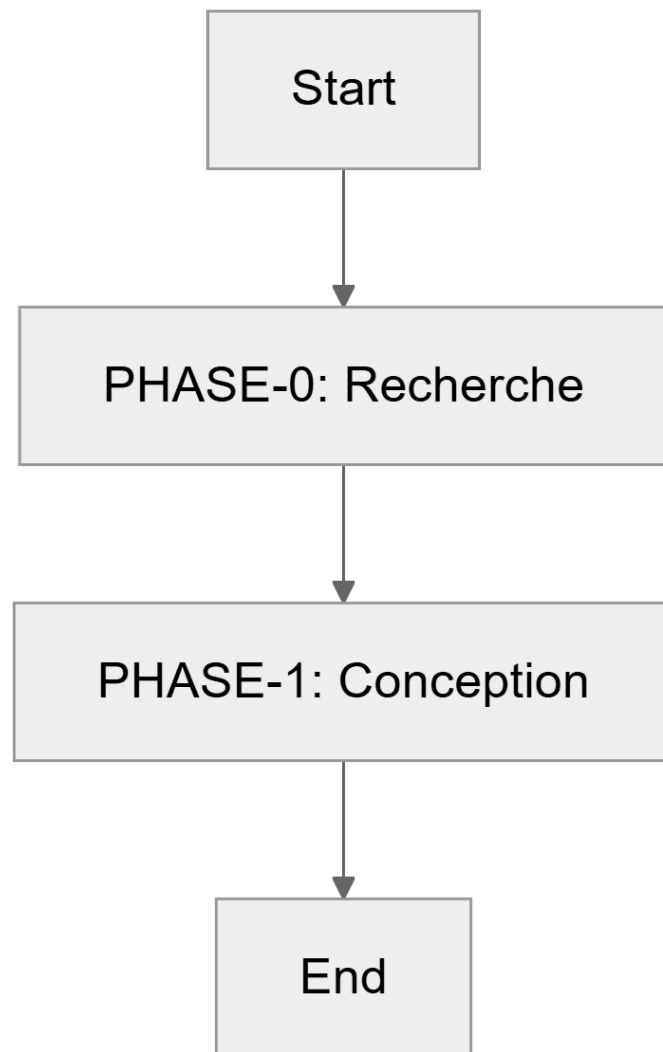
- Entity Relationship Diagram:



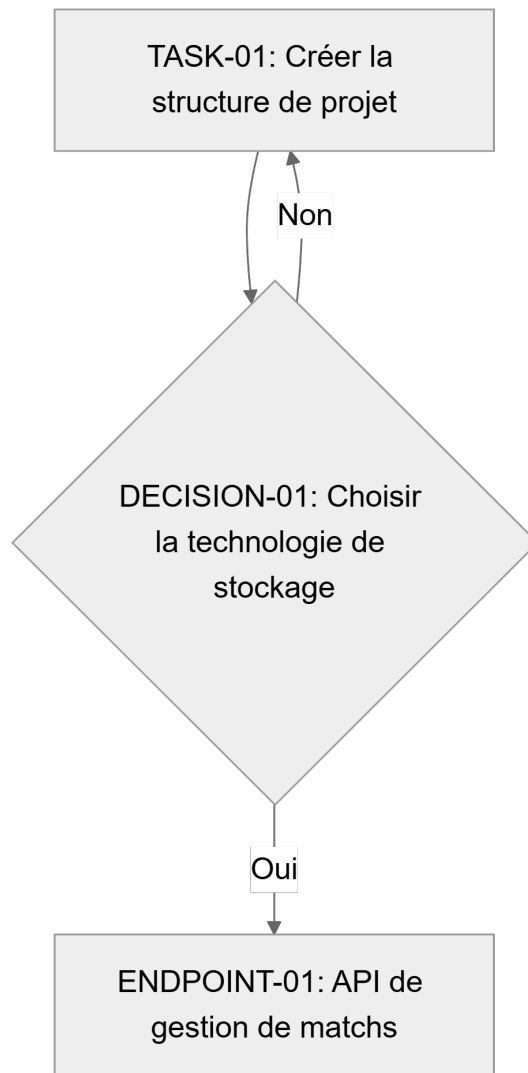
- Sequence Diagram for API Endpoint:



- Flowchart for Project Phases:



- Flowchart for Task and Decision:



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifiant	Description	Source
PHASE-0	Recherche	Implementation Plan: Football Match Manager
PHASE-1	Conception	Implementation Plan: Football Match Manager
TASK-01	Créer la structure de projet	Project Structure
DECISION-01	Choisir la technologie de stockage	Technical Context
ENDPOINT-01	API de gestion de matchs	Technical Context
ENTITY-01	Match	Technical Context

ARCH-CHOICE-01	Utiliser un modèle de données relationnel	Technical Context
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3.2 Security Rules

The system will implement the following security rules:

- Authentication and authorization using identity and access management system
- Data encryption for sensitive information
- Regular security testing and vulnerability assessment

3.3 Data Models

The system will use a relational data model to store and manage data.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

The following structural gaps have been identified:

- Missing section for data models and schemas
- Missing section for API contracts and flow
- Missing section for security and identity

4.2 Remediation & Workflow

To address these gaps, the following remediation plan will be implemented:

- Add sections for data models and schemas, API contracts and flow, and security and identity
- Review and update the technical stack and architectural constraints
- Conduct regular security testing and vulnerability assessment

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
ACTION	TECHNICAL_STACK	Technical Context	the process of executing a specific task or operation within the system
API	TECHNICAL_STACK	ENDPOINT-01	an interface that allows different applications to communicate with each other
Branch	TECHNICAL_STACK	Implementation Plan: Football Match Manager	a separate line of development in a version control system

CORS Standard	TECHNICAL_STACK	Technical Context	a security feature that restricts web pages from making requests to a different origin than the one the web page was loaded from
Constraints	BUSINESS_DOMAIN	Technical Context	limitations or restrictions on the system or its components
CoreData	TECHNICAL_STACK	Technical Context	a framework for managing model data in applications
DB	TECHNICAL_STACK	Technical Context	a repository for storing and managing data
Date	BUSINESS_DOMAIN	Implementation Plan: Football Match Manager	a point in time or a specific day
GATE	TECHNICAL_STACK	Constitution Check	a checkpoint or a control point in a process
IF	TECHNICAL_STACK	Technical Context	a conditional statement used in programming
LLVM	TECHNICAL_STACK	Technical Context	a compiler infrastructure used for building and optimizing applications
LOC	TECHNICAL_STACK	Technical Context	a measure of the size of a software program
NOT	TECHNICAL_STACK	Technical Context	a logical operator used in programming
Note	BUSINESS_DOMAIN	Implementation Plan: Football Match Manager	a comment or a remark added to a document or a plan
ONLY	TECHNICAL_STACK	Technical Context	a keyword used in programming to specify a single option or choice
Option	TECHNICAL_STACK	Technical Context	a choice or a selection available to the user
Performance Goals	BUSINESS_DOMAIN	Technical Context	targets or objectives for the performance of a system or application
Primary Dependencies	TECHNICAL_STACK	Technical Context	the main dependencies or libraries required by a project
Project Type	BUSINESS_DOMAIN	Technical Context	the type or classification of a project
Python 3.11	TECHNICAL_STACK	Technical Context	a version of the Python programming language
REMOVE	TECHNICAL_STACK	Technical Context	to delete or eliminate something

Spec	TECHNICAL_STACK	Implementation Plan: Football Match Manager	a specification or a detailed description of a system or component
Storage	TECHNICAL_STACK	Technical Context	a repository or a location for storing data
Structure Decision	BUSINESS_DOMAIN	Technical Context	a decision or a choice related to the organization or architecture of a system
Target Platform	TECHNICAL_STACK	Technical Context	the intended platform or environment for a system or application
Testing	TECHNICAL_STACK	Technical Context	the process of evaluating or verifying the functionality of a system or application
UI	TECHNICAL_STACK	Technical Context	the user interface or the visual components of a system or application
UNUSED	TECHNICAL_STACK	Technical Context	not used or utilized
WASM	TECHNICAL_STACK	Technical Context	a binary format for executing code in web browsers
iOS	TECHNICAL_STACK	Technical Context	a mobile operating system developed by Apple